

Section I**Multiple Choice 10 Marks**

Use the multiple choice answer sheet for Questions 1 – 10

1. The solution of the inequation $1 - 3x \leq -x$ is

- A. $x < \frac{1}{2}$ B. $x \geq -\frac{1}{2}$ C. $x \geq \frac{1}{2}$ D. $x < -\frac{1}{2}$

2. $\sqrt{x^5}$ can be written as

- A. $x^{\frac{2}{5}}$ B. $x^{\frac{5}{2}}$ C. $\sqrt[5]{x^2}$ D. $x^{5.2}$

3. 0.000006053 in scientific notation is

- A. 6.053×10^{-6} B. 6053×10^4 C. 6.053×10^6 D. 6053×10^{-6}

4. Factorising $x^3 - 27y^3$ becomes

- A. $(x - 3y)(x^2 + 3xy + 9y^2)$ B. $(x - 3y)^3$
B. C. $(x^2 - 3)(x + 9y^2)$ D. $(x^2 - 3y^2)(x + 9y)$

5. $(3\sqrt{7} + 2\sqrt{3})(3\sqrt{7} - 2\sqrt{3})$ is equal to

- A. 51 B. 75 C. $51 + 4\sqrt{3}$ D. 63

6. The gradient function of $f(x) = \sqrt{x}$ is

A. $\frac{1}{2}x$

B. $\frac{2}{\sqrt{x}}$

C. $\frac{\sqrt{x}}{2}$

D. $\frac{1}{2\sqrt{x}}$

7. $1 + \frac{2}{a+b} =$

A. $\frac{a+b+2}{a+b}$

B. $\frac{3}{a+b}$

C. 3

D. $\frac{2}{a+b}$

8. $\frac{2 - \tan 60^\circ}{\sec^2 45^\circ}$ in exact form is

A. $2\sqrt{3}$

B. $1 - \frac{\sqrt{3}}{2}$

C. $-\frac{1}{\sqrt{3}}$

D. $\frac{2 - \sqrt{3}}{\sqrt{2}}$

9. The number of sides of a regular polygon with exterior angles of 30° is

A. 6

B. 8

C. 10

D. 12

10. The equation of the line that passes through (2, -5) and is parallel to the line $2x - 3y + 1 = 0$ can be expressed as

A. $2x + 3y - 19 = 0$

B. $2x - 3y - 19 = 0$

C. $3x + 2y + 4 = 0$

D. $3x + 2y + 4 = 0$

Section II**Attempt Questions 11 to 16**

Answer each question in separate writing booklets.

All necessary working should be shown in every question

Question 11	14 Marks	Start a new booklet	Marks
(a) Simplify	$2\sqrt{75} - 3\sqrt{48}$		2
(b) Solve	$x^2 = 5x$		2
(c) Evaluate, correct to 3 significant figures	$\frac{6.2^5 - 5.2^4}{\sqrt{18} - 4 \times 6^{\frac{1}{3}}}$		2
(d) Solve	$ 2x + 1 \geq 5$ and graph your solution on the number line		3
(e) Simplify	$\frac{2}{m^2 - 4} - \frac{3}{m + 2}$		3
(f) Solve simultaneously	$\begin{aligned} 5p - t &= -28 \\ 16p - 5t &= -14 \end{aligned}$		2

Question 12 13 Marks**Start a new booklet****Marks**

(a) Sketch the following, showing all essential features

8

(i) $2x - 4y - 1 = 0$

(ii) $y = \sqrt{16 - x^2}$

(iii) $y = -\frac{4}{1+x}$

(iv) $y = x^3 + 8$

(b) Shade the region in the Cartesian plane for which

3 $y < x - 2$, $y \geq 0$ and $x \geq 6$ hold simultaneously.

(c) Find the centre and radius of the circle

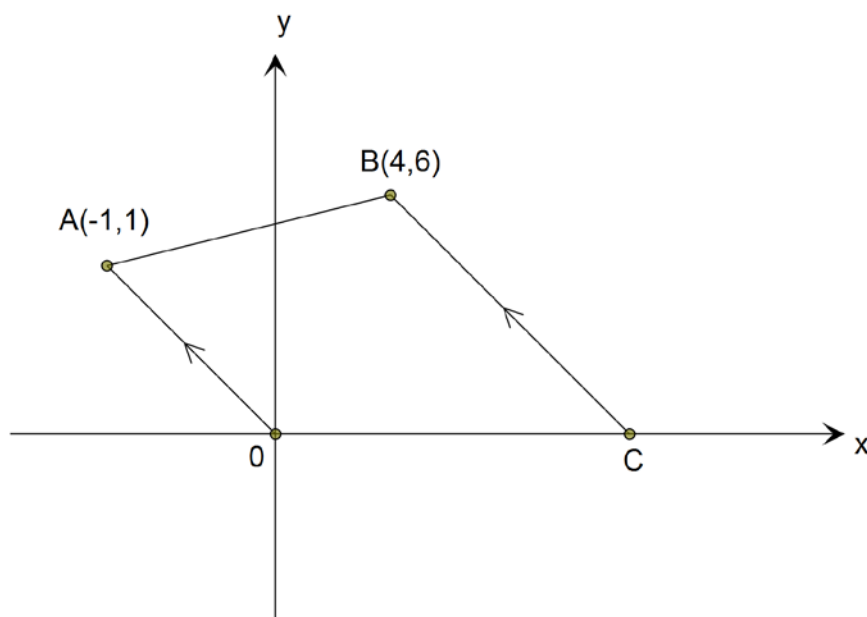
2

$x^2 - 2x + y^2 + 10y + 22 = 0$.

Question 13 13 Marks

Start a new booklet

Marks



In the diagram, OABC is a trapezium with $OA \parallel CB$. The coordinates of O, A and B are $(0, 0)$, $(-1, 1)$ and $(4, 6)$ respectively.

- | | |
|--|---|
| (i) Calculate the length of OA. | 2 |
| (ii) Write down the gradient of the line OA. | 1 |
| (iii) What is the size of $\angle AOC$? | 1 |
| (iv) Find the equation of the line BC, and hence find the coordinates of C. | 3 |
| (v) Show that the perpendicular distance from O to the line BC is $5\sqrt{2}$. | 2 |
| (vi) Hence or otherwise, calculate the area of the trapezium OABC. | 2 |
| (vii) Find the equation of the line that passes through O and is perpendicular to the line BC. | 2 |

Question 14 16 Marks

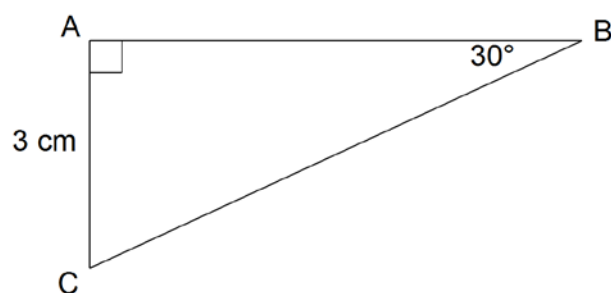
Start a new booklet

Marks

(a) Find $\cos \theta$ if $\operatorname{cosec} \theta = 4$ for $90^\circ \leq \theta \leq 180^\circ$ 2

(b) Sketch $y = \sin x$ in the domain $0^\circ \leq x \leq 360^\circ$. Show all essential features. 1

(c) Find the exact length of AB. 2



(d) Solve each equation for $0^\circ \leq \theta \leq 360^\circ$. Correct answers to the nearest minute.

(i) $\cos \theta = -\frac{3}{4}$ 2

(ii) $\cot \theta = 4$ 2

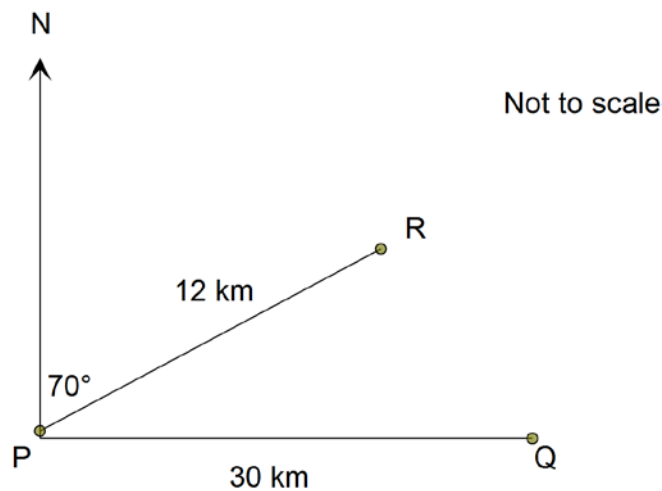
(e) Find the exact value of $\sin 225^\circ$ 1

Question 14 continues on next page.

Question 14 continued

Marks

(f)

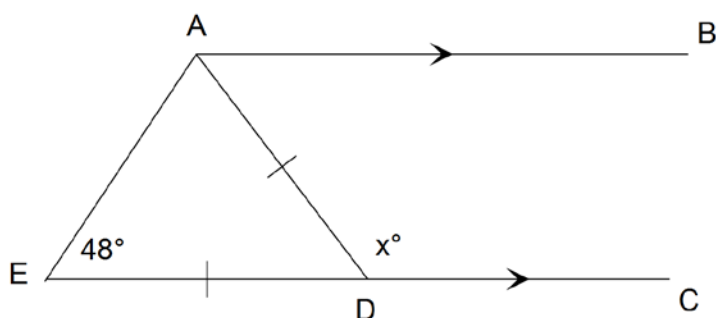


The diagram shows a point P which is 30 km due west of the point Q. The point R is 12 km from P and has a bearing from P of 070° .

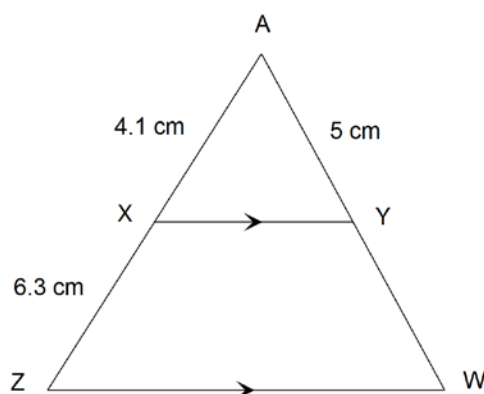
(i) Find the distance of R from Q, correct to 1 decimal place. 2

(ii) Find the bearing of R from Q, correcting your answer to the nearest degree. 2

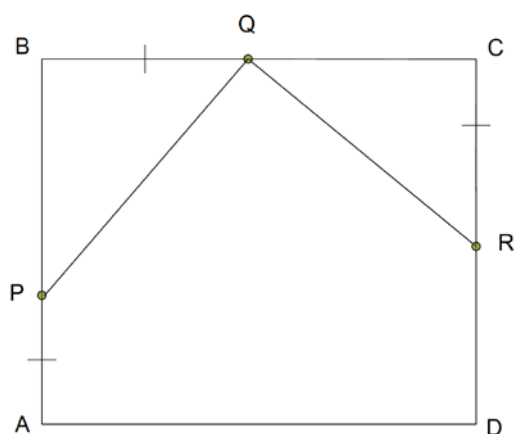
(g) Prove $\sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$ 2

Question 15 13 Marks**Start a new booklet****Marks**(a) Find the value of x , giving reasons.**2**

(b) Find the length of YW, correct to 1 decimal place.

2

(c)



Not to scale

In the diagram, ABCD is a square. The points P, Q and R lie on AB, BC and CD respectively, such that $AP = BQ = CR$.

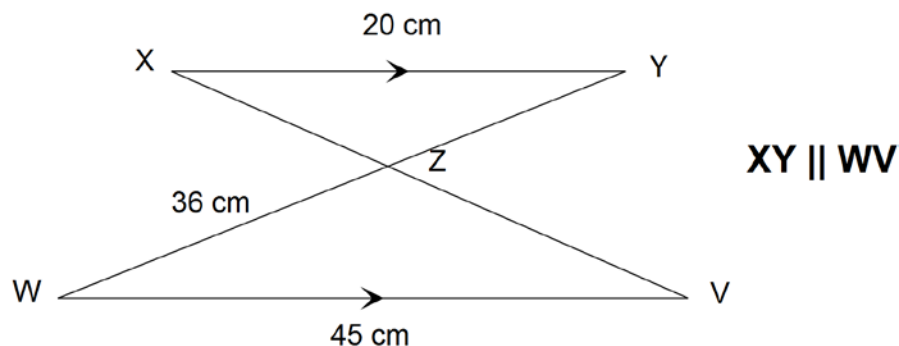
(i) Prove the triangles PBQ and QCR are congruent

3(ii) Prove $\angle PQR$ is a right angle**2***Question 15 continues on next page.*

Question 15 continued

Marks

(d) Given the following diagram,



- (i) Prove $\triangle XYZ \cong \triangle VWZ$ 3
- (ii) Hence find the length of ZY 1

QUESTION 16 21 Marks

Start a new booklet

Marks

(a) Differentiate the following with respect to x

- (i) $y = 3 - 2x^2 - 7x^4$ 2
- (ii) $y = 5\sqrt{x} - \frac{x^2}{2}$ 2
- (iii) $y = (1 - 2x)(1 + 3x)$ 2
- (iv) $y = \frac{4x - 1}{1 - x}$ 2
- (v) $y = (8x^2 + 4x)^3$ 2
- (vi) $y = 2x\sqrt[3]{x - 5}$ leave your answer without a fractional or negative index 2
- (vii) $y = (3x - 1)(2 - 3x)^4$ 2

Question 16 continues on next page.

QUESTION 16 continued**Marks**

- (b) Differentiate $f(x) = x^2 - 2x$ from 1st principles and find the gradient of the tangent to the curve at $x = 1$. **3**
- (c) Differentiate $y = x^2 + bx + c$ and hence find b and c given that the line $3x + y - 5 = 0$ is a normal to the curve at the point $(3, -1)$ **3**

END OF PAPER

SOLUTIONS

2013

SECTION I (10 MARKS)

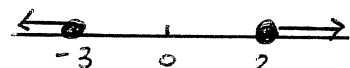
1. $x \geq \frac{1}{2}$ C
2. $x^{\frac{5}{2}}$ B
3. 6.053×10^{-6} A
4. $(x-3y)(x^2+3xy+9y^2)$
A
5. 51 A
6. $\frac{1}{2\sqrt{x}}$ D
7. $\frac{a+b+2}{a+b}$ A
8. $1 - \frac{\sqrt{3}}{2}$ B
9. $n=12$ D
10. $2x-3y-19=0$ B.

SECTION IIQUESTION 11. (14 MARKS)

$$\begin{aligned}
 (a) \quad & 2\sqrt{25/3} - 3\sqrt{16/3} \quad \perp \\
 & = 10\sqrt{3} - 12\sqrt{3} \\
 & = -2\sqrt{3} \quad \perp
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & x^2 - 5x = 0 \\
 & x(x-5) = 0 \\
 \therefore & x=0 \text{ or } x=5 \\
 & \quad \quad \quad \perp \quad \quad \quad \perp
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & -2786.056774... \quad \checkmark \\
 & = -2790 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad & 2x+1 \geq 5 \quad \text{or} \quad -(2x+1) \geq 5 \\
 & 2x \geq 4 \quad \quad \quad -2x-1 \geq 5 \\
 & x \geq 2 \quad \quad \quad -2x \geq 6 \\
 & \quad \quad \quad \quad \quad x \leq -3 \quad \checkmark
 \end{aligned}$$


$$\begin{aligned}
 (e) \quad & \frac{2}{(m+2)(m-2)} - \frac{3(m-2)}{(m+2)(m-2)} \quad \perp \\
 & = \frac{2 - 3m + 6}{(m+2)(m-2)} \quad \perp \\
 & = \frac{8 - 3m}{(m+2)(m-2)} \quad \perp
 \end{aligned}$$

$$\begin{aligned}
 (f) \quad & t = 5p + 28 \quad \text{--- ①} \\
 & 16p - 5t = -14 \quad \text{--- ②}
 \end{aligned}$$

Sub ① into ②

$$\begin{aligned}
 16p - 5(5p + 28) &= -14 \\
 16p - 25p - 140 &= -14 \\
 -9p &= 126
 \end{aligned}$$

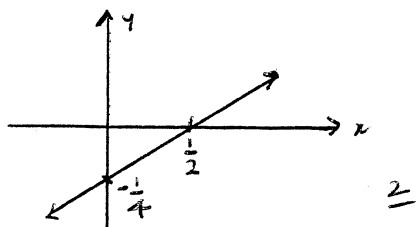
$$p = -14 \quad \perp$$

$$\begin{aligned}
 t &= 5x - 14 + 28 \\
 &= -42 \\
 & \quad \quad \quad \perp
 \end{aligned}$$

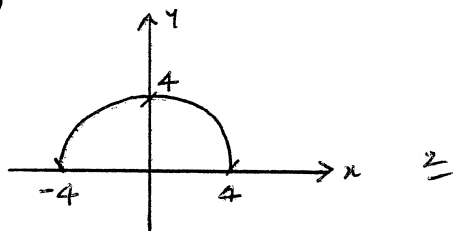
QUESTION 12 (13 MARKS)

(a)

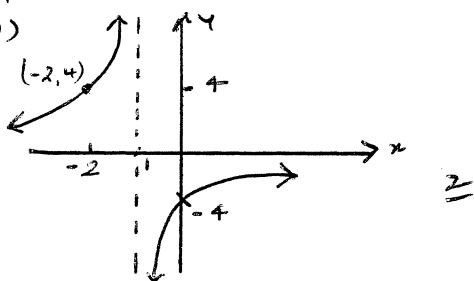
(i)



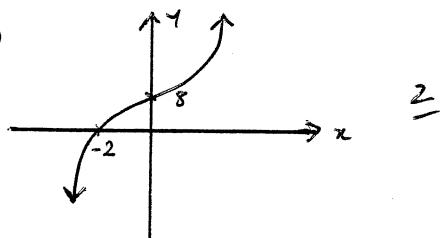
(ii)



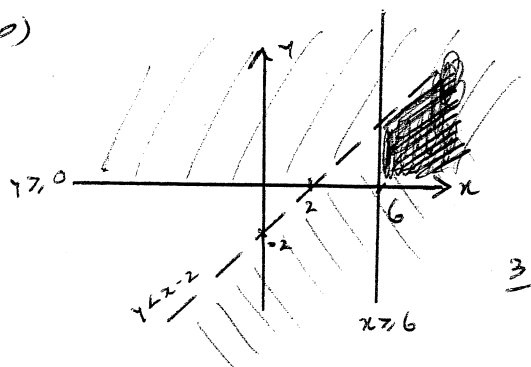
(iii)



(iv)



(b)



$$(c) \quad x^2 - 2x + y^2 + 10y = -22$$

$$x^2 - 2x + 1 + y^2 + 10y + 25 = -22 + 1 + 25$$

$$(x-1)^2 + (y+5)^2 = 4$$

Centre $(1, -5)$ Radius $= 2$

QUESTION 13 (13 MARKS)

$$(i) \quad d_{OA} = \sqrt{1^2 + 1^2} = \sqrt{2} = 1.41 \text{ (2 dp)}$$

$$(ii) \quad m = -1$$

$$(iii) \quad \tan \theta = -1$$

$$\theta = 135^\circ$$

$$\therefore \angle AOC = 135^\circ$$

$$(iv) \quad m = -1 \quad (4, 6)$$

$$y - 6 = -1(x - 4)$$

$$y - 6 = -x + 4$$

$$y = -x + 10$$

$$\text{Let } y = 0 \quad x = 10$$

$\therefore$ words of C are $(10, 0)$

$$(v) \quad x + y - 10 = 0 \quad (0, 0)$$

$$d_{\perp} = \frac{|1 \times 0 + 1 \times 0 - 10|}{\sqrt{1^2 + 1^2}}$$

$$= \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{10\sqrt{2}}{2} = 5\sqrt{2} \text{ units}$$

$$(vi) \quad A = \frac{1}{2} \times 5\sqrt{2} (\sqrt{2} + 6\sqrt{2})$$

$$= \frac{1}{2} \times 5\sqrt{2} \times 7\sqrt{2}$$

$$= \frac{1}{2} \times 35 \times 2$$

$$= 35 \text{ sq. units}$$

$$(vii) \quad \text{grad } BC = -\frac{6}{6} = -1$$

$$\therefore \text{Grad of line } \perp BC \text{ is } 1$$

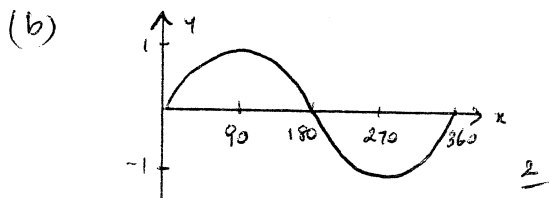
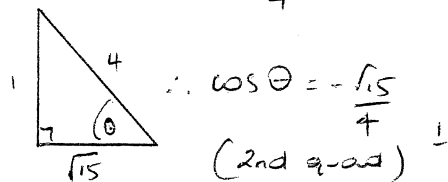
Eqn through $(0, 0)$ grad 1

$$y - 0 = 1(x - 0)$$

$$y = x$$

QUESTION 14 (16 MARKS)

(a) $\operatorname{cosec} \theta = 4$
 $\frac{1}{\sin \theta} = 4$
 $\sin \theta = \frac{1}{4}$ 1



(c) $\tan 30^\circ = \frac{3}{AB}$ 1

$AB = 3 \div \tan 30^\circ$
 $= 3 \div \frac{1}{\sqrt{3}}$
 $= 3\sqrt{3} \text{ cm}$ 1

(d) (i) $\cos \theta = -\frac{3}{4}$
 Basic angle: $\theta = 41^\circ 25'$

2nd Quad: $138^\circ 35'$ 1
 3rd Quad: $221^\circ 25'$ 1

(ii) $\frac{1}{\tan \theta} = 4$

$\tan \theta = \frac{1}{4}$

1st Quad: $\theta = 14^\circ 2'$ ✓
 3rd Quad: $\theta = 194^\circ 2'$ ✓

(e) $\sin 225^\circ = \sin (180 + 45)$
 $= -\sin 45^\circ$
 $= -\frac{1}{\sqrt{2}}$
 $= -\frac{\sqrt{2}}{2}$ 1

(f) (i) $RQ^2 = 12^2 + 30^2 - 2 \cdot 12 \cdot 30 \cdot \cos 20^\circ$
 $RQ = \sqrt{12^2 + 30^2 - 2 \cdot 12 \cdot 30 \cdot \cos 20^\circ}$
 $= 19.168$
 $\approx 19.2 \text{ cm}$ 1

(ii) $\frac{\sin \theta}{12} = \frac{\sin 20^\circ}{19.2}$

$\sin \theta = \frac{12 \sin 20^\circ}{19.2}$
 $= 0.213$ 1

$\theta = 12^\circ 21'$

Bearing is $270^\circ + 12^\circ 21'$
 $= 282^\circ 21'$ 1

(or) $\frac{\sin \theta}{12} = \frac{\sin 20^\circ}{RQ}$
 $\sin \theta = 0.21$

$\theta = 12^\circ 22' \therefore \text{Bearing}$
 $282^\circ 22'$

(g) $\text{LHS} = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$ 1

$= \frac{1 + \sin \theta}{\cos \theta}$ 1

$= \text{RHS}$

QUESTION 15 (13 MARKS)

(a) $\angle ADE = 48^\circ$ (base $\angle$ s isos Δ) ✓

$\therefore x = 132^\circ$ (suppl. $\angle$ EDC) ✓

(b) $\frac{YW}{5} = \frac{6.3}{4.1}$ ✓

$YW = \frac{5 \times 6.3}{4.1}$
 $= 7.6829 \dots$
 $= 7.7 \text{ cm}$ ✓

(c) (i) $\angle PBQ = \angle RCQ$ (right $\angle$ s in square)
 $BQ = CR$ (given) ✓

$BP = QC$ (since $AB = BC$, $BQ = PC$) ✓

$\therefore \Delta PBQ \equiv \Delta QCR$ (SAS) ✓

(ii) $\angle BPQ + \angle BQP = 90^\circ$ ($\angle$ sum $\Delta = 180^\circ$)

$\angle RQC + \angle CQR = 90^\circ$ ($\angle$ sum $\Delta = 180^\circ$) ✓

$\angle BQP + \angle CQR + \angle PQR = 180^\circ$
 $90^\circ + \angle PQR = 180^\circ$

$\therefore \angle PQR = 90^\circ$ ✓

(d) (i) $\angle YXZ = \angle WVZ$ (alt $\angle$ s; $XY \parallel WV$) ✓

$\angle ZWV = \angle XYZ$ (alt $\angle$ s; $XY \parallel WV$) ✓

$\angle XZY = \angle VZW$ (vert. opp $\angle$ s) ✓

$\therefore \Delta XYZ \parallel \Delta VWZ$ (equiangular) ✓

(ii) $\frac{ZY}{36} = \frac{20}{45}$

$ZY = \frac{20}{45} \times 36$

$= 16 \text{ cm}$ ✓

QUESTION 16 (21 MARKS)

(a) (i) $y' = -4x - 28x^3$ ✓

(ii) $y = 5x^{\frac{1}{2}} - \frac{x^2}{2}$

$y' = \frac{5}{2} x^{-\frac{1}{2}} - x$ ✓

$= \frac{5}{2\sqrt{x}} - x$ ✓

(iii) $y' = (1-2x) \cdot 3 + (1+3x) \cdot -2$
 $= 3 - 6x - 2 - 6x$
 $= 1 - 12x$ ✓

(iv) $y' = \frac{(1-x) \cdot 4 - (4x-1) \cdot -1}{(1-x)^2}$
 $= \frac{4 - 4x + 4x - 1}{(1-x)^2}$
 $= \frac{3}{(1-x)^2}$ ✓

(v) $y' = 3(8x^2 + 4x)^2 (16x + 4)$
 $= 3(16x + 4)(8x^2 + 4x)^2$ ✓

(vi) $y = 2x(x-5)^{\frac{1}{3}}$

$y' = 2x \left[\frac{1}{3}(x-5)^{-\frac{2}{3}} \cdot 1 \right] + (x-5)^{\frac{1}{3}} \cdot 2$
 $= \frac{2x}{3} (x-5)^{-\frac{2}{3}} + 2(x-5)^{\frac{1}{3}}$

$= 2(x-5)^{\frac{1}{3}} \left(\frac{x(x-5)^{-1}}{3} + 1 \right)$

$= 2\sqrt[3]{x-5} \left(\frac{x}{3(x-5)} + 1 \right)$ ✓

(accept $y' = \frac{2x}{3\sqrt[3]{(x-5)^2}} + 2\sqrt[3]{x-5}$)

$$\begin{aligned}
 \text{(vii)} \quad y' &= (3x-1) \left[4(2-3x)^3 \cdot -3 \right] + (2-3x)^4 \cdot 3 \\
 &= -12(3x-1)(2-3x)^3 + 3(2-3x)^4 \quad \underline{1} \\
 &= -3(2-3x)^3 (4(3x-1) + 2-3x) \\
 &= -3(2-3x)^3 (12x-4+2-3x) \\
 &= -3(2-3x)^3 (9x-2) \quad \underline{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad f(x) &= x^2 - 2x \\
 f(x+h) &= (x+h)^2 - 2(x+h) \\
 &= x^2 + 2xh + h^2 - 2x - 2h
 \end{aligned}$$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - (x^2 - 2x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 2h}{h} \\
 &= \lim_{h \rightarrow 0} 2x + h - 2 \\
 &= 2x - 2 \quad \underline{2}
 \end{aligned}$$

Gradient at $x=1$ is $y' = 2 \times 1 - 2 = 0$.

$$\text{(c)} \quad y = x^2 + bx + c$$

$$\frac{dy}{dx} = 2x + b \quad \underline{1}$$

Line $y = -3x + 5$ has grad -3

$\therefore$ Grad of normal is $\frac{1}{3}$

use $(3, -1)$ and $\frac{dy}{dx} = \frac{1}{3}$

$$\left(\begin{array}{l} \text{Sub} \\ \text{into} \\ y' = 2x + b \end{array} \right) \frac{1}{3} = 2 \times 3 + b$$

$$\underline{b} = -5 \frac{2}{3} \quad \underline{1}$$

$$\text{Sub into original} \quad -1 = 3^2 - 5 \frac{2}{3} \times 3 + c$$

$$\therefore \underline{c} = 7 \quad \underline{1}$$